

DAVID L. GIBBS

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Professional Summary

Senior Research Scientist with 10+ years in AI/ML for computational pathology developing custom deep learning models for digital histopathology, spatial biomarker discovery, and multi-modal data integration in immuno-oncology. Deep expertise in foundation model fine-tuning and adaptation, including current work developing multiclass classifiers on pathology foundation models for tumor microenvironment characterization. Proven track record translating imaging-based AI into biological insights through cross-functional collaborations with pathologists and clinicians.

Key Skills

AI/ML & Computational Imaging

- Technical team management
 - Foundation Model Fine-tuning & Adaptation
 - Deep Learning for Digital Pathology (CNNs, U-Net architectures, CycleGAN)
 - Multi-task Learning & Multi-headed Architectures
 - Whole Slide Image Analysis & Spatial Statistics
 - Foundation Model Fine-tuning & Transfer Learning
 - Computer Vision & Image Segmentation
 - Biomarker Discovery & Patient Stratification
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Professional Experience

Institute for Systems Biology (ISB) – Senior Research Scientist – Seattle, WA, 2019-Present

Computational Imaging & AI/ML Development

- Co-Principal Investigator on CRDC Imaging Data Commons (36TB+ repository), architecting scalable infrastructure for pathology and radiology data enabling thousands of cancer researchers.
- Leading development of multi-headed classifier built on Virchow2 foundation model for NCI Cancer Systems Biology Consortium, simultaneously predicting cancer subtypes, stromal architecture, and immune cell populations (T cells, macrophage subtypes) from H&E histopathology (manuscript in preparation).

- Developed and deployed CycleGAN-based deep learning models for IHC-to-H&E stain transformation in whole slide images, achieving successful nuclear structure preservation across tissue types.
- Built custom spatial analysis pipelines for digital pathology methods to quantify lymphocyte clustering patterns in tumor microenvironments related to treatment response.

Cross-functional Research Leadership & Biomarker Discovery

- Co-led computational component of CRUK Grand Challenge studying tissue inflammation and cancer progression, partnering with cancer biologists, pathologists, and clinicians across 7 institutions.
- Key member of TCGA PanCancer Atlas Immune Response working group, developing immune gene signatures that identified novel cancer-immune subtypes validated across 11,000+ patient samples (5700+ citations).
- Co-author and R developer on CRI iAtlas, an interactive web portal enabling researchers to explore immunotherapy biomarkers across multiple data modalities.
- Developed and established metadata standards for imaging and spatial data types in Human Tumor Atlas Network (HTAN).

Mentorship & Scientific Communication

- Published 39 peer-reviewed papers with h-index of 19.
- Teach research methods courses in the summer intern program.
- Mentor junior scientists and students on AI/ML projects.

Key Technical Products

- gssnng: Single cell gene set scoring tool integrated into the scverse ecosystem (scanpy), enabling robust analysis of single-cell spatial data.
- Spatial Analysis Framework: Developed multi-scale pyramid approaches for analyzing tissue architecture using spatial autocorrelation statistics.
- tbccsi Python AI Prediction package: Built ML pipeline for cell type classification from H&E pathology images.

Education

- Postdoc Fellow, Institute for Systems Biology, 2016-2017, Advisor: Dr. Ilya Schumilovich
- Ph.D. & Postdoc, Bioinformatics & Computational Biology, Oregon Health and Science University, 2009-2015, Advisor: Dr. Shannon McWeeney
- Bachelor of Science, Chemistry, Reed College, 1999

Select Publications

- *Gibbs et al. (2025)* Robust Cluster Prediction Across Data Types Validates Association of Sex and Therapy Response in GBM. *Cancers*.
 - *Strasser, Gibbs, et al. (2025)* Concerted epithelial and stromal changes during progression of Barrett's Esophagus to invasive adenocarcinoma exposed by multi-scale, multi-omics analysis. *Dev Cell*.
 - *Thorsson & Gibbs et al. (2019)*. The Immune Landscape of Cancer. *Immunity*.
 - *Gibbs & Shmulevich (2017)*. Influence Maximization in Biological Networks. *PLoS Comput Biol*.
 - *Gibbs et al. (2014)* Multi-omic network signatures of disease. *Frontiers in genetics*
 - h-index = 19
 - <https://scholar.google.com/citations?user=IMvBJ3IAAAAJ>
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Honors & Appointments

- 2023 Innovator Award at ISB
 - Co-PI of Imaging Data Commons, a central hub for cancer research data
 - Postdoctoral Fellow, Institute for Systems Biology
 - National Library of Medicine Postdoctoral Fellowship
 - ISCB Travel Fellowship
 - NLM Predoctoral Fellowship (ARRA Grant)
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Technical Tools & Platforms

Deep Learning & AI

- Frameworks: PyTorch (primary), TensorFlow, MONAI
- Foundation Models: Virchow2, pathology-specific vision transformers
- Computer Vision: OpenCV, Scikit-Image, PIL, whole slide image processing libraries
- ML Tools: Scikit-learn, multi-task learning, transfer learning, domain adaptation

Cloud & Computing

- Platforms: Google Cloud Platform, BigQuery, distributed computing
- Containerization: Docker (if true)
- Version Control: Git/GitHub

Programming & Data Science

- Languages: Python (expert), R, SQL, C++11
- Bioinformatics: Scanpy, Bioconductor, spatial analysis tools, single-cell methods
- Data Processing: Pandas, NumPy, image processing pipelines